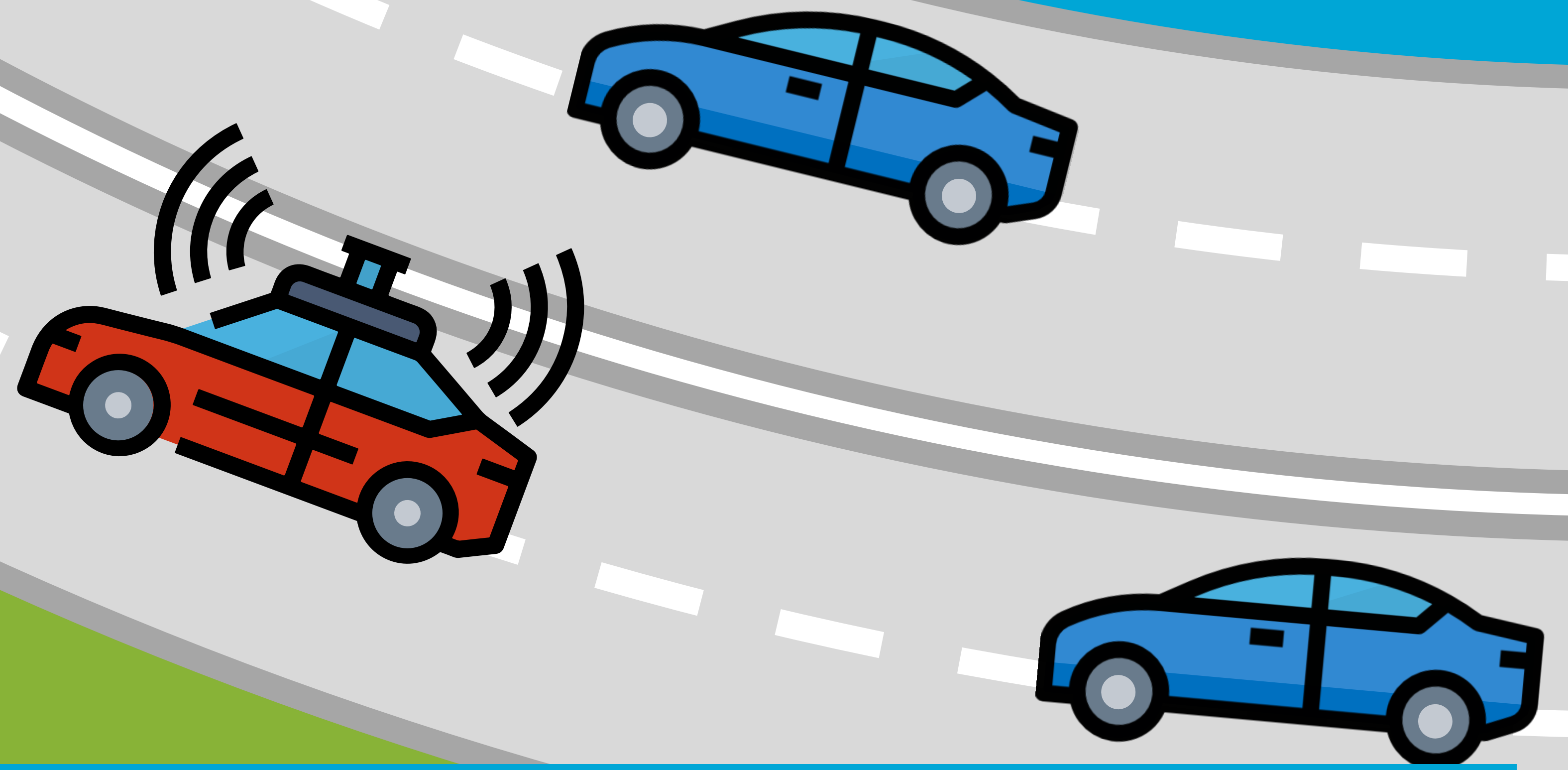




Learning collision risk proactively from naturalistic driving data at scale

THE AUTHORS

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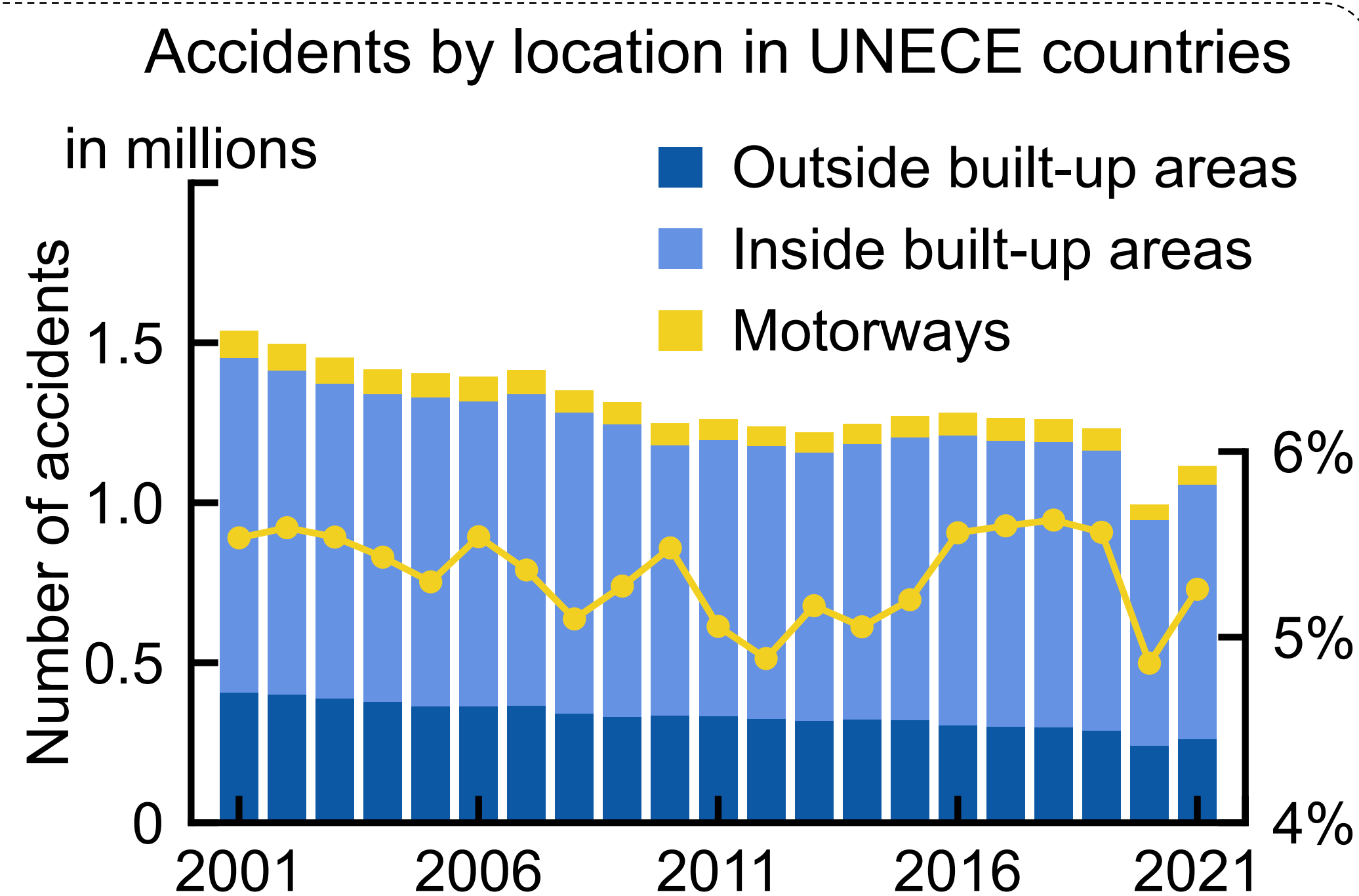
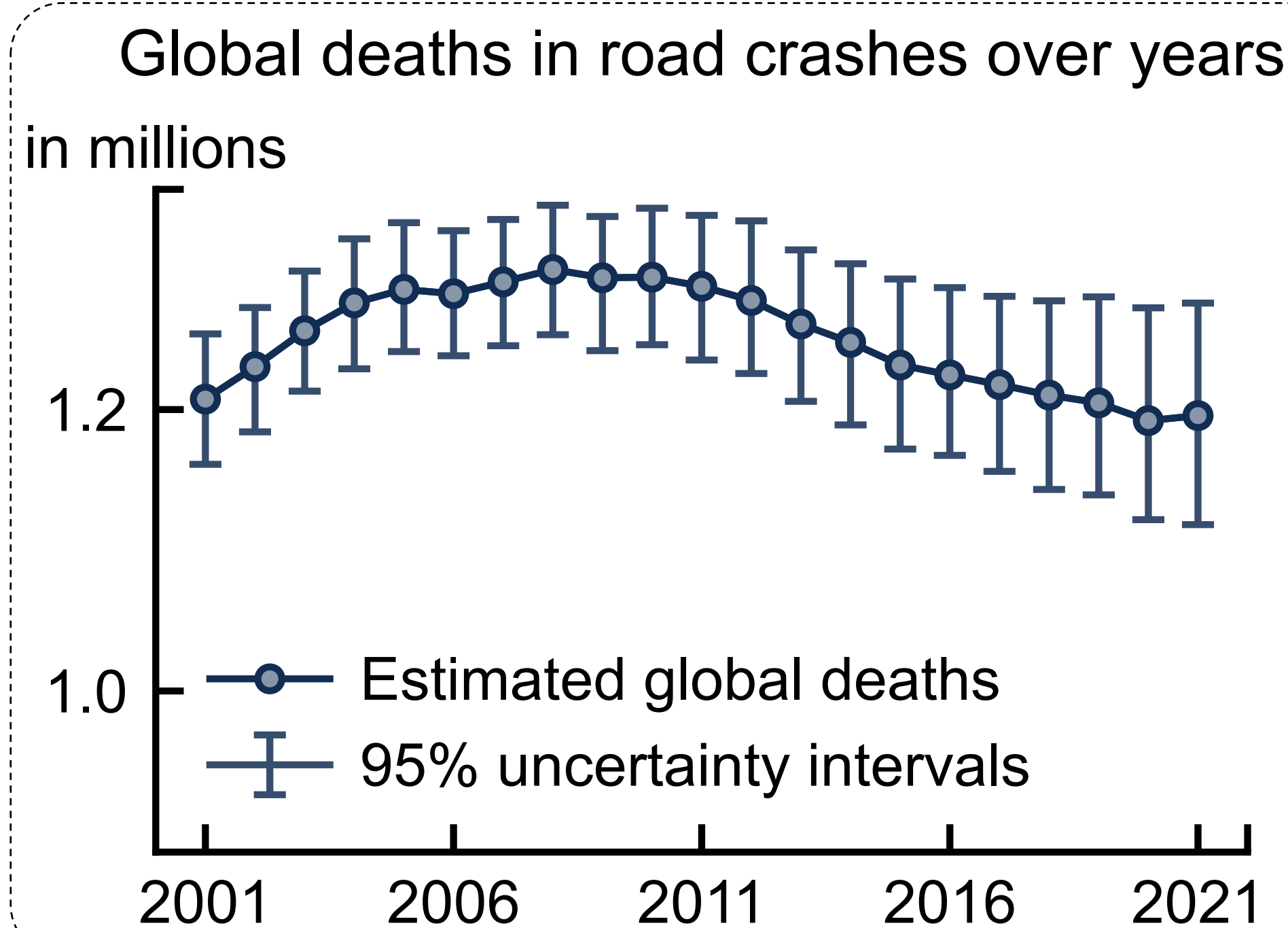
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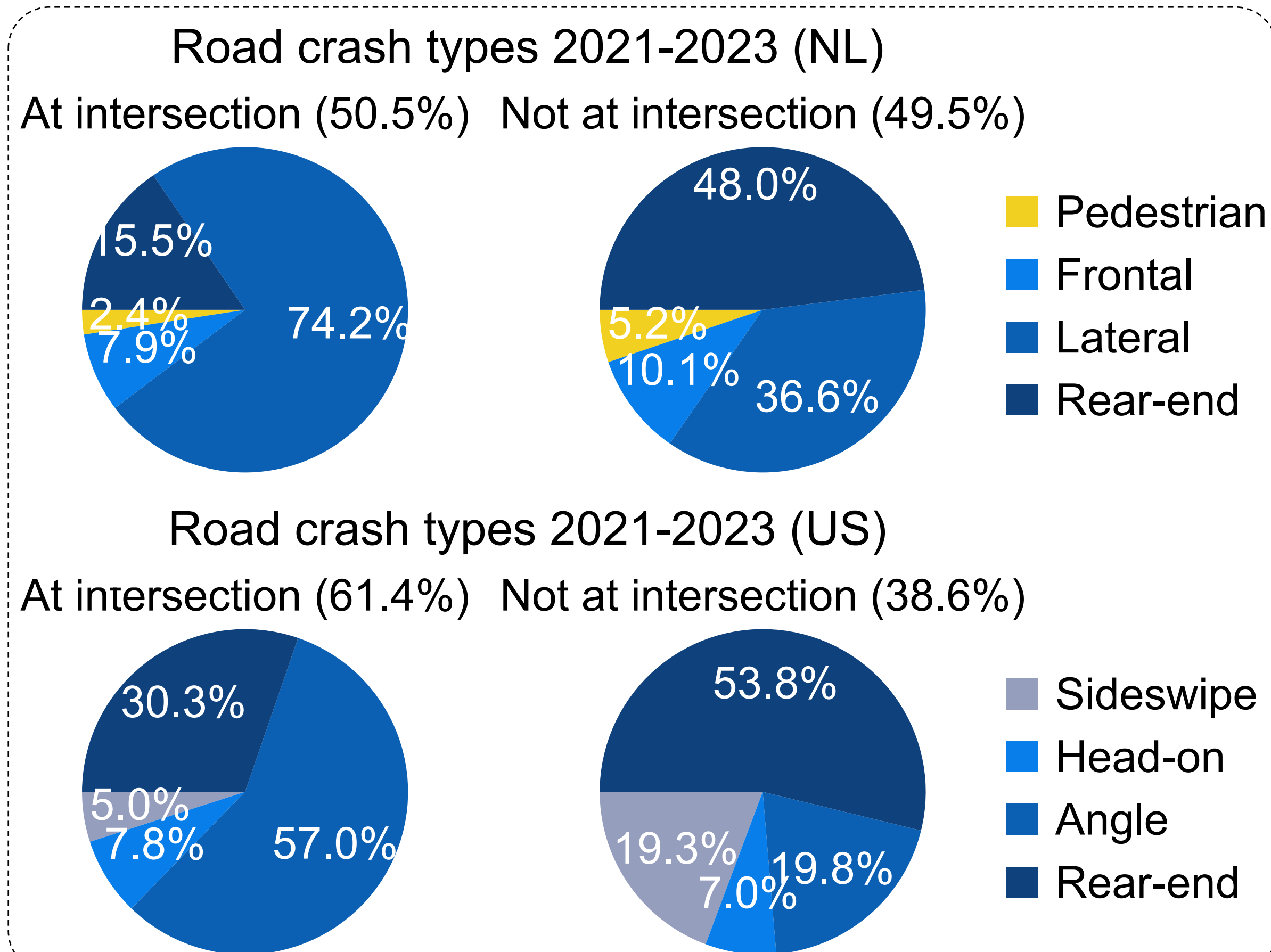
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Accurately and proactively alerting drivers or automated systems to emerging collisions is crucial for road safety, particularly in highly interactive and complex urban environments. Existing methods either require labour-intensive annotation of sparse risk, struggle to consider varying contextual factors, or are tailored to limited scenarios. Here we present the Generalised Surrogate Safety Measure (GSSM), a data-driven approach that learns collision risk from naturalistic driving without the need for crash or risk labels.

BACKGROUND



- Over 1 million deaths and tens of millions of injuries occur annually due to road traffic accidents.
- Urban areas are hotspots, accounting for over 67% of the accidents.
- More than half road user collisions happen at intersections and most are during lateral interactions (merging, turning, crossing, etc.).
- Collisions are not sudden events but the result of escalating conflicts, securing the crucial time window for intervention is important.



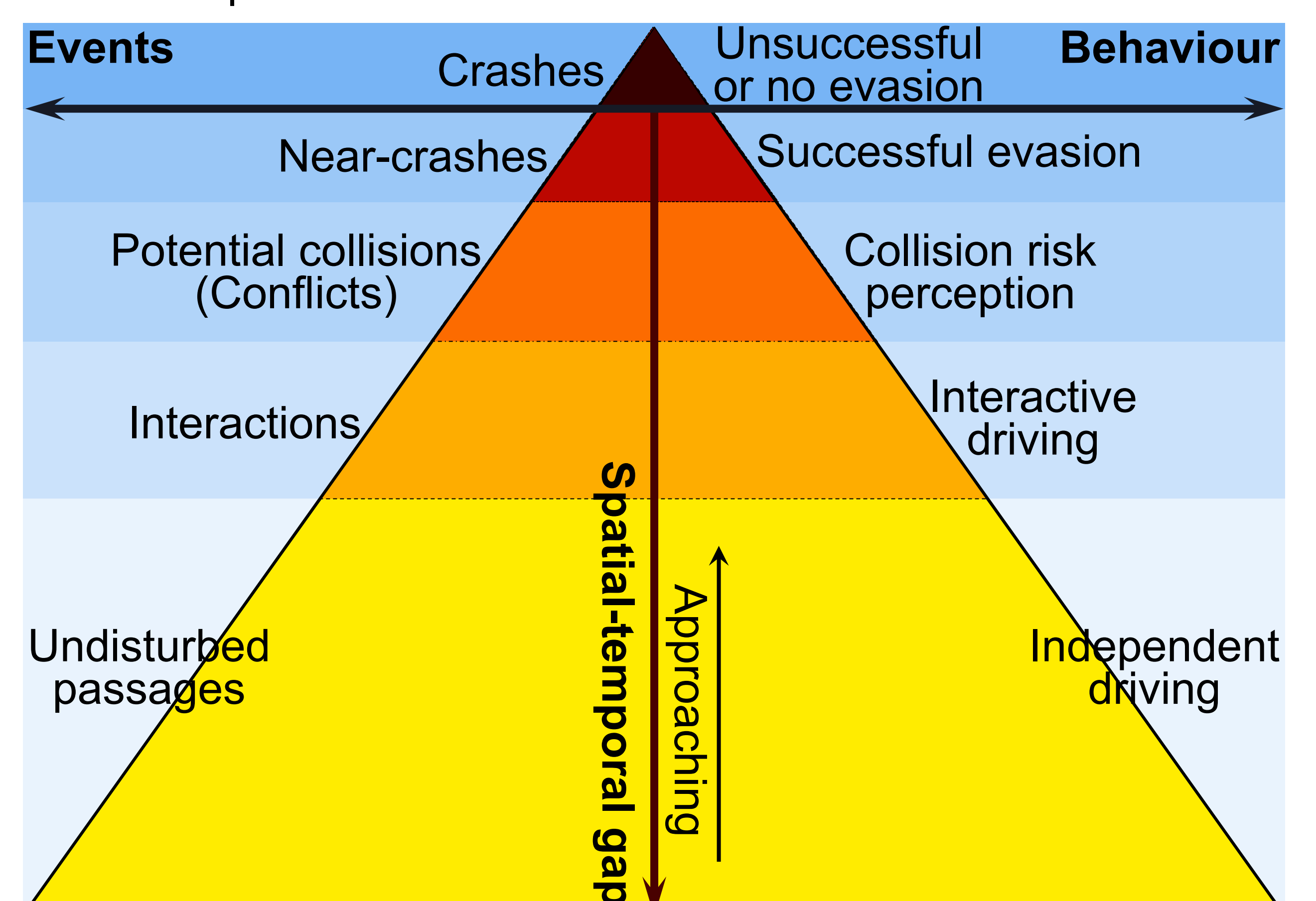
RESEARCH OBJECTIVE

Develop a **context-aware, generalisable, and scalable** methodology to quantify the collision risk in multi-directional traffic interactions.

- Context-aware:** account for all information that can characterise interaction;
- Generalisable:** cover diverse traffic interactions and handle never-seen contexts;
- Scalable:** leverage everyday observations, instead of relying on factual crashes.

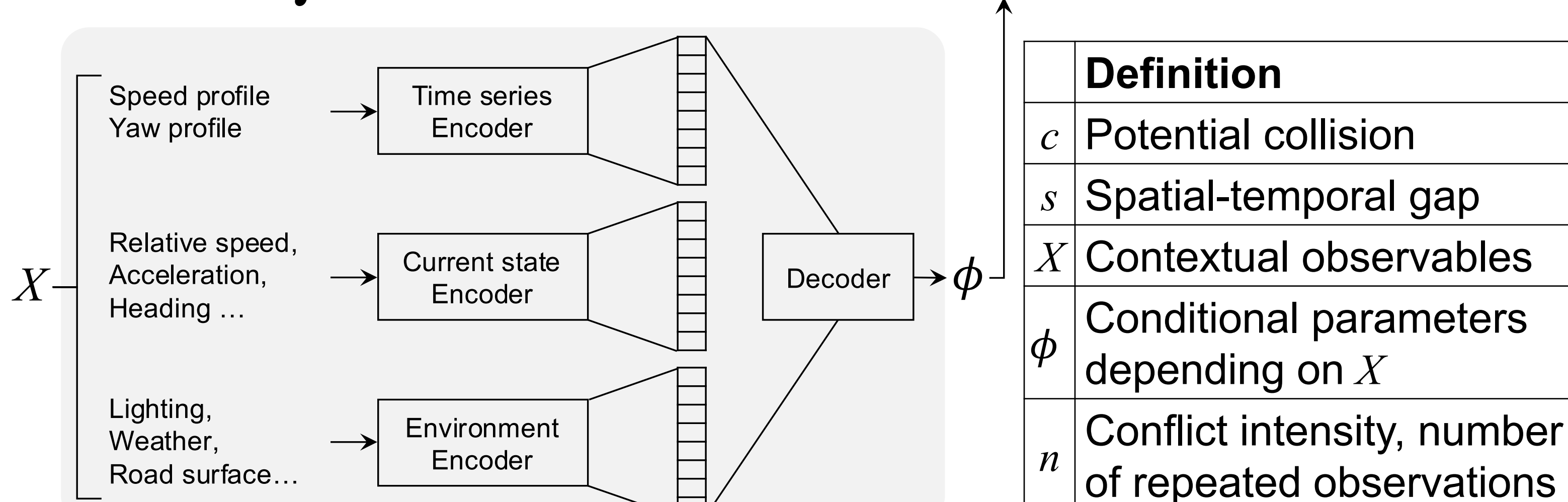
METHODOLOGY IN A NUTSHELL

GSSM captures the **patterns of normal driving** and estimates the extent to which a traffic interaction **deviates from typical safe behaviour towards an unsafe state** in the interaction context, outputting a continuous risk level and the corresponding likelihood of a potential collision.

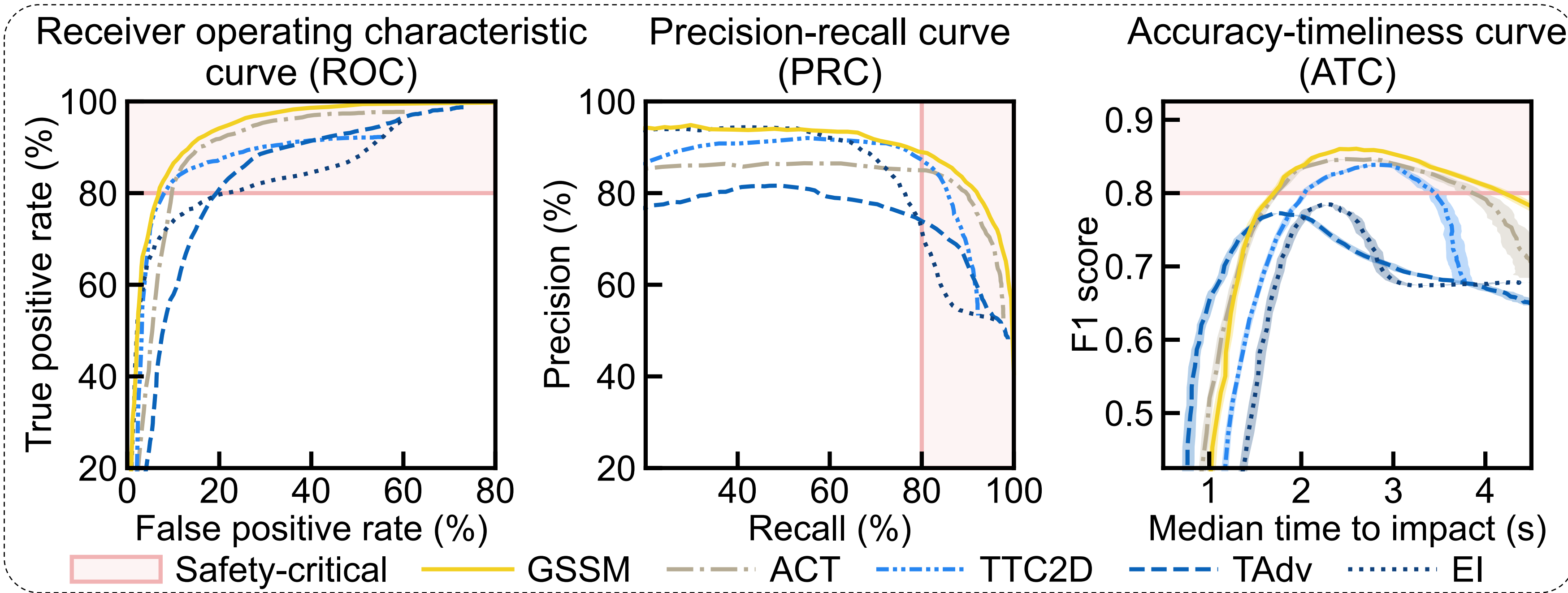


FRAMEWORK

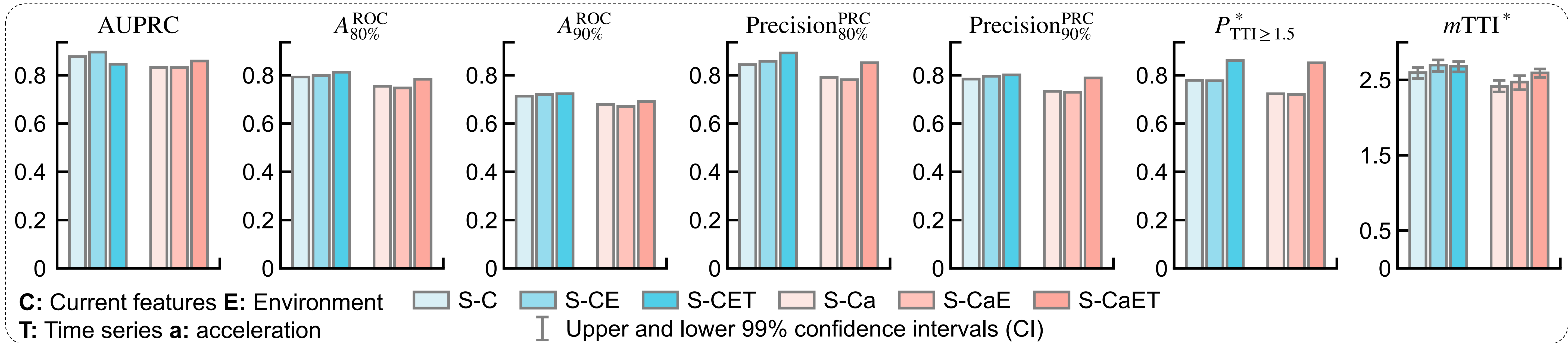
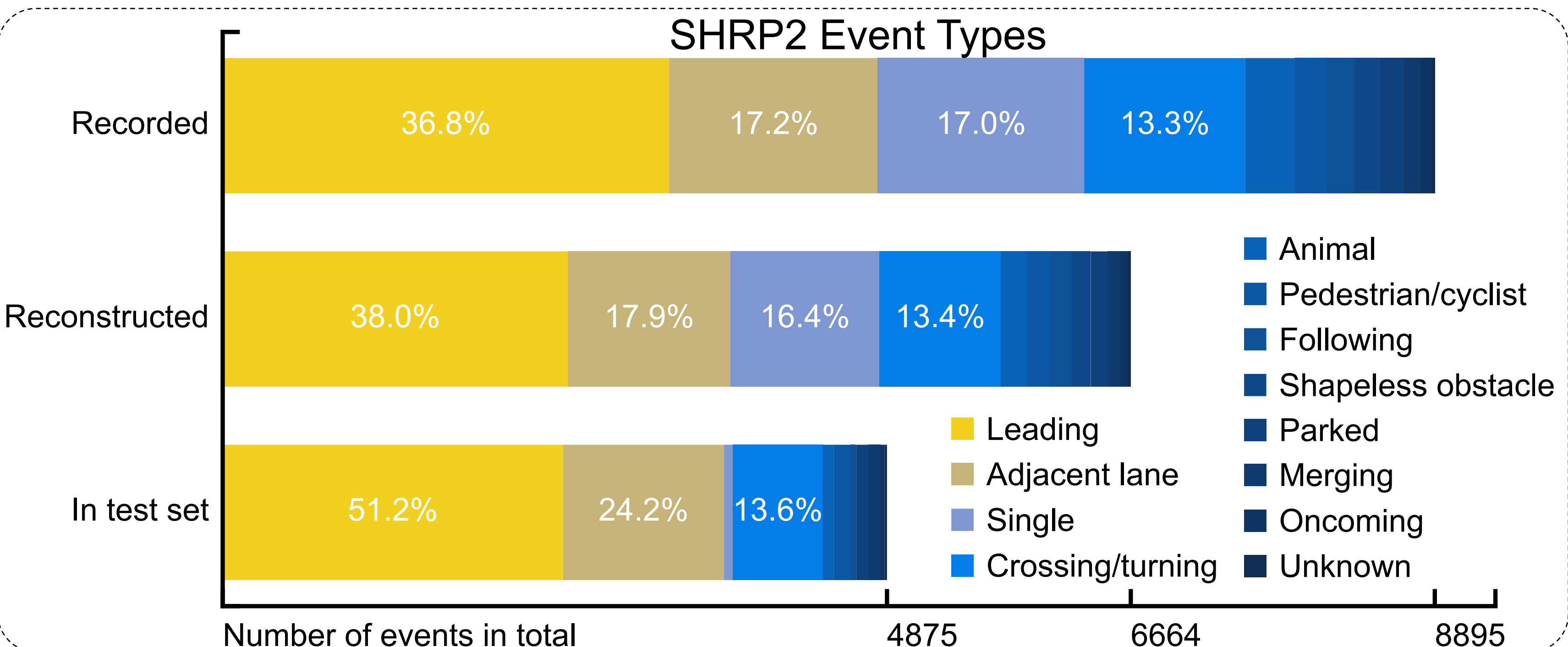
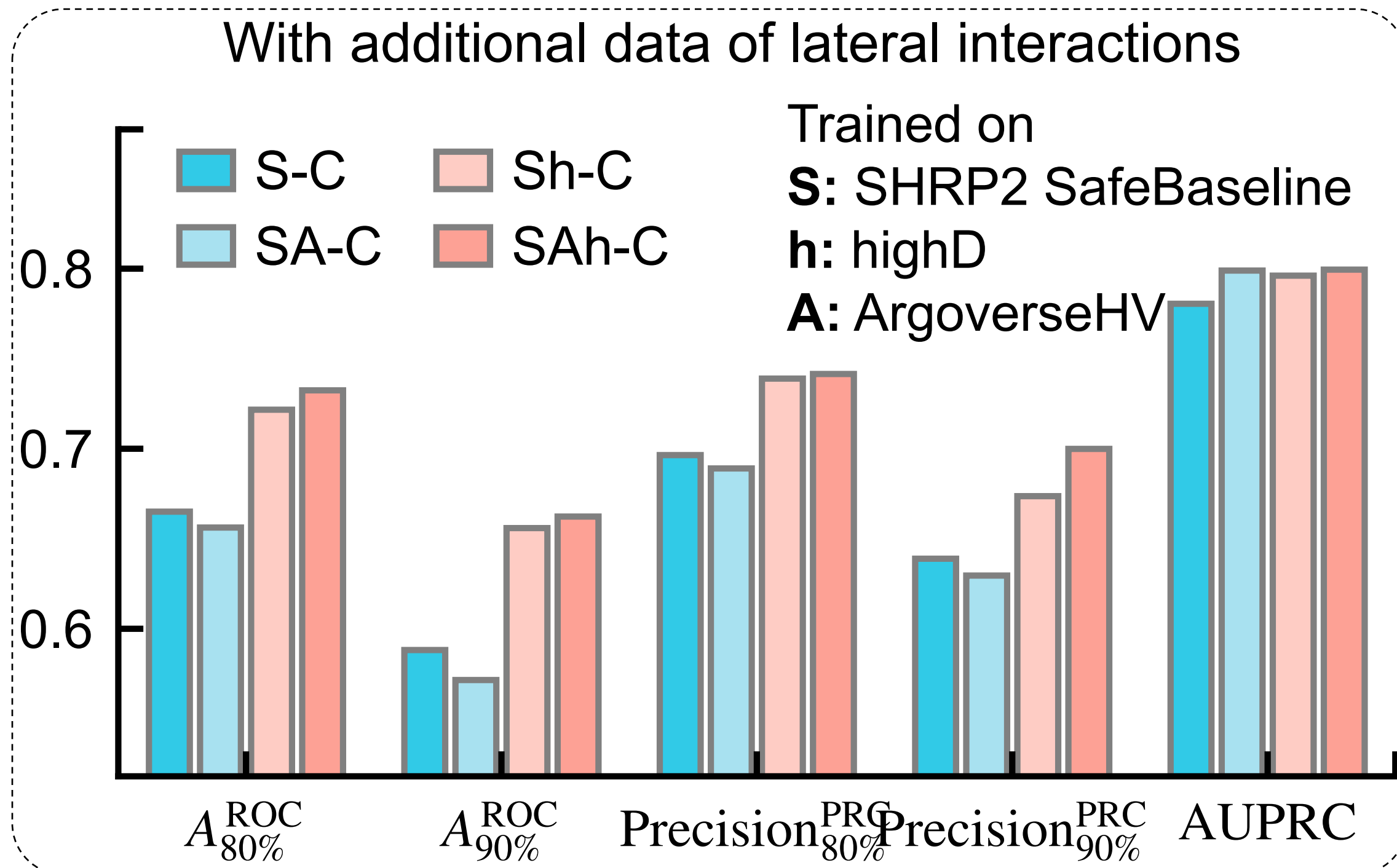
$$p(c | s, X) = \int p(c | s, \phi) p(\phi | X) d\phi = p(c | s, \phi(X)) = [\Pr(S > s | \phi(X))]^n$$



RESULTS



Eval. Metric	Definition
AUPRC	Area under PRC
A_{R}^{ROC}	Area under the ROC curve when true positive rate is larger than R
$Precision_{R}^{PRC}$	Highest Precision on the PRC when Recall is larger than R
$P_{TTI \geq 1.5}^*$	Percentage of events with time-to-impact (TTI) ≥ 1.5 s among correctly detected (near-)crashes
$mTTI^*$ [Q1, Q3]; 99%CI	Median TTI across all correctly detected (near-)crashes. Q1 and Q3 are the first and third quartiles. 99%CI is the 99% confidence interval of $mTTI$



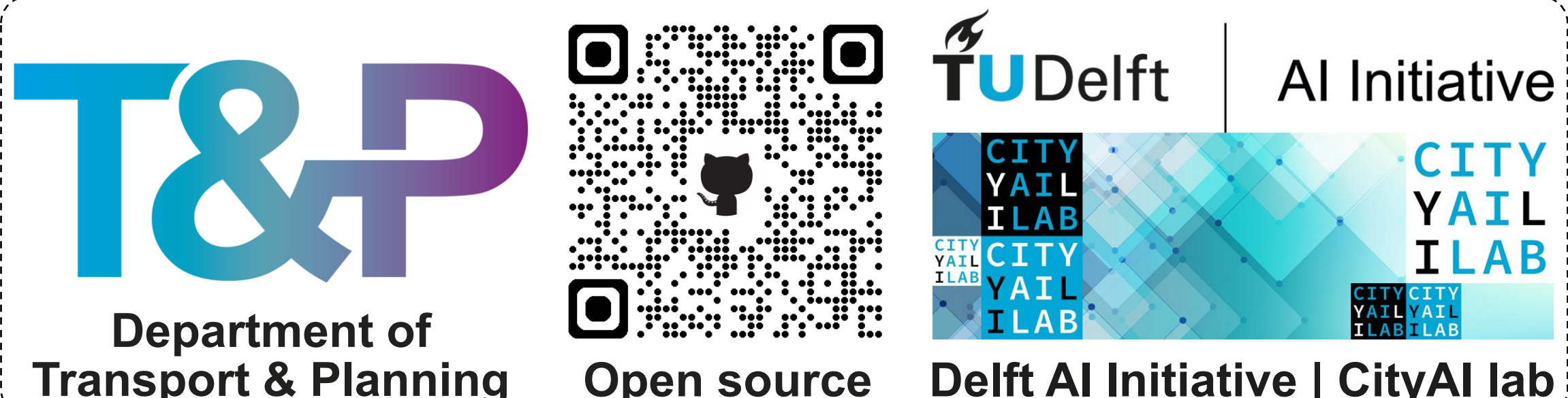
Event	Number of events	Method	AUPRC	$A_{80\%}^{ROC}$	$A_{90\%}^{ROC}$	$Precision_{80\%}^{PRC}$	$Precision_{90\%}^{PRC}$	$P_{TTI \geq 1.5}^*$	$mTTI^*$ [Q1, Q3]; 99%CI
Rear-end	1787	GSSM	0.86	0.93	0.89	0.95	0.93	0.88	2.53 [1.92, 3.45]; 2.45--2.58
		ACT	0.87	0.85	0.79	0.88	0.88	0.72	2.28 [1.30, 3.09]; 2.19--2.34
		TTC2D	0.87	0.70	0.45	0.93	0.91	0.86	2.49 [1.85, 3.26]; 2.43--2.56
		TAdv	0.73	0.73	0.64	0.80	0.75	0.49	1.44 [0.76, 2.04]; 1.39--1.50
		EI	0.91	0.75	0.56	0.93	0.81	0.73	2.10 [1.45, 2.88]; 2.02--2.16
Adjacent lane	611	GSSM	0.77	0.69	0.61	0.70	0.66	0.83	3.38 [1.88, 5.15]; 3.19--3.61
		ACT	0.69	0.53	0.33	0.68	0.61	0.90	4.67 [2.88, 6.14]; 4.40--4.94
		TTC2D	0.70	0.32	0.00	0.63	N/A	0.85	3.68 [2.06, 5.38]; 3.36--3.85
		TAdv	0.59	0.38	0.27	0.51	0.48	0.75	2.38 [1.45, 3.34]; 2.19--2.51
		EI	0.64	0.22	0.04	0.45	0.45	0.85	3.48 [1.93, 5.50]; 3.33--3.74
Crossing/turning	93	GSSM	0.75	0.60	0.53	0.76	0.73	0.97	4.85 [3.63, 6.38]; 4.21--5.46
		ACT	0.74	0.35	0.02	0.79	0.68	0.93	5.16 [3.48, 6.46]; 4.62--5.40
		TTC2D	0.68	0.00	0.00	N/A	N/A	0.95	4.77 [3.21, 6.33]; 4.32--5.31
		TAdv	0.71	0.37	0.18	0.73	0.65	0.82	2.90 [2.04, 4.07]; 2.39--3.29
		EI	0.69	0.00	0.00	N/A	N/A	0.93	4.13 [2.66, 5.71]; 3.30--4.70
Merging	29	GSSM	0.87	0.80	0.72	0.83	0.77	0.89	5.11 [3.23, 7.30]; 3.58--5.60
		ACT	0.71	0.52	0.23	0.76	0.66	1.00	5.17 [3.75, 6.80]; 4.52--6.49
		TTC2D	0.79	0.00	0.00	N/A	N/A	0.86	3.83 [2.17, 4.85]; 2.66--4.23
		TAdv	0.68	0.25	0.00	0.52	N/A	0.73	2.71 [1.32, 3.87]; 1.52--3.39
		EI	0.78	0.29	0.00	0.55	N/A	0.90	3.21 [2.12, 4.85]; 2.52--4.43
With pedestrian/cyclist /animal	31	GSSM	0.85	0.38	0.31	0.80	0.80	1.00	5.94 [4.89, 7.16]; 4.89--7.16
		ACT	0.71	0.07	0.00	0.76	N/A	0.97	5.38 [3.90, 6.54]; 4.20--6.09
		TTC2D	0.70	0.00	0.00	N/A	N/A	0.90	4.51 [2.31, 5.57]; 2.82--5.14
		TAdv	0.71	0.09	0.01	0.71	0.70	0.81	3.19 [1.76, 5.34]; 2.39--4.62
		EI	0.73	0.00	0.00	N/A	N/A	0.87	3.78 [2.48, 5.55]; 2.55--5.36

CONCLUSION

A basic GSSM using only instantaneous motion kinematics achieves an **area under the precision-recall curve of 0.9**, and secures a **median time advance of 2.6 seconds** to prevent potential collisions. **Incorporating additional interaction patterns and contextual factors provides further performance gains**. Across interaction scenarios such as rear-end, merging, and turning, **GSSM consistently outperforms existing baselines in accuracy and timeliness**. These results establish GSSM as a scalable, context-aware, and generalisable foundation to identify risky interactions before they become unavoidable, supporting proactive safety in autonomous driving systems and traffic incident management.

FUTURE APPLICATIONS AND IMPLICATIONS

- Traffic management
- Next-Generation Vehicles
- Real-Time Road Safety
- Accelerated Safety Policy



This poster is based on the following article that more details are referred to: Yiru Jiao, Simeon C. Calvert, Sander van Cranenburgh, and Hans van Lint. (2026). Learning collision risk proactively from naturalistic driving data at scale. Nature Machine Intelligence (in press). Available at [arXiv: 2505.13556](https://arxiv.org/abs/2505.13556).